

	Case Name: CRONOGRAMA DE OBRA Project Owner: Eder Omar Vilca Carrillo	Sector	Construction	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
			One of nine std. scenarios	
Submitted by	Matilde Casado and Margarida Antunes		User defined distributions	
Date			Automatic tracking	
File Name	C2025-28	Project Control	Tracking based on user input	

1. Project description

Project authenticity

This project improves the Puncu-Cocha-Lluncu irrigation system through reservoir repair, drainage, and geomembrane installation, boosting water efficiency and sustainability. The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	58
Planned Duration (PD)	60 days
Budget At Completion (BAC)	156771.1 €
Renewable Resources	8
Consumable Resources	-

* standard eight-hour 7 days of the week

Network topology	
Serial/Parallel (SP)	26%
Activity Distribution (AD)	37%
Length of Arcs (LA)	0%
Topological Float (TF)	47%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	100	0	N/A
CRI-tau	100	0	N/A

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	0	0	N/A
CRI-tau	0	0	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	5.9	14.7	5.1
SI	19.4	25.3	1.2
SSI	5.1	11.8	2.4
CRI-r	9.8	8.6	1.0
CRI-rho	10.2	8.4	1.2
CRI-tau	10.6	7.2	0.4

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	10.9	-4.6	1	0.0	0.0
PV - SPI	16.3	0.8	CPI	0.0	0.0
PV - SCI	16.3	0.8	SPI	6.1	4.9
ED - 1	9.5	-7.3	SPI(t)	7.7	5.0
ED - SPI	16.3	0.8	SCI	6.1	4.9
ED - SCI	16.3	0.8	SCI(t)	7.7	5.0
ES - 1	14.9	-12.1	0.8 CPI + 0.2 SPI	1.7	1.5
ES - SPI(t)	27.7	-7.1	0.8 CPI + 0.2 SPI(t)	2.3	1.9
ES - SCI(t)	27.7	-7.1			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 1 tracking period. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0.0	0.0	-16.0	1.0	1.0	1.0	1.0
std dev	N/A	N/A	N/A	N/A	N/A	N/A	N/A
final	0.0	0.0	-16.0	1.0	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	60 days	Real Duration	92 days	Late	53.3%
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EAC(t)	Real Accuracy			
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	89.0	N/A	3.3	3.3
PV - SPI	90.0	N/A	2.2	2.2
PV - SCI	91.0	N/A	1.1	1.1
ED - 1	92.0	N/A	0.0	0.0
ED - SPI	93.0	N/A	1.1	-1.1
ED - SCI	94.0	N/A	2.2	-2.2
ES - 1	95.0	N/A	3.3	-3.3
ES - SPI(t)	96.0	N/A	4.3	-4.3
ES - SCI(t)	97.0	N/A	5.4	-5.4

2.3.3.3. Cost forecasting

BAC	156771.1 €	Real Cost	156771.1 €	Under Budget	0%
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EAC	Real Accuracy			
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	0.0	N/A	0.0	0.0
CPI	0.0	N/A	0.0	0.0
SPI	0.0	N/A	0.0	0.0
SPI(t)	0.0	N/A	0.0	0.0
SCI	0.0	N/A	0.0	0.0
SCI(t)	0.0	N/A	0.0	0.0
0.8 CPI + 0.2 SPI	0.0	N/A	0.0	0.0
0.8 CPI + 0.2 SPI(t)	0.0	N/A	0.0	0.0